Analysis of Mach-Zehnder Interferometers Made Using Photonic Integrated Circuit Technology

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***Abstract*—In this paper, I present several imbalanced Mach-Zehnder interferometers (MZI) made using photonic integrated circuit (PIC) technology. MZIs split incident light equally into two waveguides, and then recombine the light after some distance. If the two waveguides have different lengths, an interference pattern will appear in the transmission spectrum. The distance between minima in the interference pattern is known as the free spectral range, and it is related to the group index of the waveguide.**

I designed four different MZI circuits using three different length mismatches: 60 µm, 120 µm and 180 µm. By fitting an MZI model to the measured transmission spectrum, I extracted the group indices of the waveguides. The extracted group indices were between 4.12 and 4.17. This is lower than the expected value of 4.20, which was found through finite-difference eigenmode simulations. This discrepancy could be due to the waveguide bends, or due to a slight fabrication variation. For example, for one of the MZI circuits, if I assume that the length mismatch is 1% smaller than the intended value, the extracted group velocity changes from 4.17 to 4.20.

I also present several other PIC designs, including circuits using Bragg gratings, Fabry-Pérot interferometers, photonic crystals and ring resonators.

I. Introduction

PHOTONIC integrated circuits (PICs) leverage traditional integrated circuit fabrication technology to build high-density optical circuits. In particular, silicon PICs use CMOS-compatible fabrication to make silicon waveguides and optical devices. Due to the high refractive index of silicon, these circuits can be very compact. PICs are now being used in a variety of different applications, including high-speed data transmission, medical sensing and quantum computing.

Silicon PICs are typically made using silicon-on-insulator (SOI) wafers. These wafers are commonly used in high-performance electronics as a result of the lower capacitance and leakage current offered by the buried oxide layer. However, they also offer an excellent platform for optical waveguides. A common SOI wafer composition for silicon photonics is (from top to bottom): 220 nm thick silicon, $2-3$ µm thick silicon dioxide (also known as the *buried oxide layer*), and then the thick silicon handle.

To create an optical waveguide, we need a channel made from a material with a high refractive index surrounded by a material with a low refractive index. For SOI-based PICs, the waveguides are made in the top silicon layer ($n_{\mathrm{Si}} = \sqrt{\varepsilon_{\mathrm{r,Si}}} \approx 3.47$ where ε_{r} is the relative permittivity), which sits on top of the buried oxide layer ($n_{\mathrm{SiO_2}} = \sqrt{\varepsilon_{\mathrm{r,SiO_2}}} \approx 1.44$). After etching the waveguide, a top cladding layer is often deposited. The net result is a silicon channel completely surrounded by silicon dioxide. The approximate dielectric properties of silicon (Si) and silicon dioxide (SiO_2) are summarized in Table I [1]. Note that SiO_2 has a much weaker dependence on temperature and wavelength.

TABLE I
DIELECTRIC PROPERTIES OF SILICON AND SILICON OXIDE [1].

Property	Si	SiO_2
n	3.47	1.44
ε_{r}	12.0	2.07
$\mathrm{d}n/\mathrm{d}\lambda$	$-7.6 \times 10^{-5}\ \mathrm{nm}^{-1}$	$-1.2 \times 10^{-5}\ \mathrm{nm}^{-1}$
$\mathrm{d}n/\mathrm{d}T$	$1.87 \times 10^{-4}\ \mathrm{K}^{-1}$	$8.5 \times 10^{-6}\ \mathrm{K}^{-1}$

In this report, I present a few different silicon PIC designs, including imbalanced Mach-Zehdner interferometers, Bragg gratings, Fabry-Pérot interferometers, photonic crystals and ring resonators. The primary objective was to measure the group index of the silicon waveguides using an imbalanced Mach-Zehdner interferometers. This was achieved by measuring the interference pattern present in the transmission data. The distance between minima is known as the free spectral range, from which I can calculate the effective group index. The other circuits are interesting and they were used to learn more about silicon photonics.

This report was written in partial fulfilment of the requirements for the *UBCx: Silicon Photonics Design, Fabrication, and Data Analysis* course taught by Prof. Lukas Chrostowski at the University of British Columbia (UBC), Vancouver, Canada.

II. Modelling Strip Waveguides

For my first PIC design, I kept the waveguide parameters very simple in order to achieve at least one working circuit. I chose to base my design on the TE polarization with a waveguide height of 220 nm and a width of 500 nm. I simulated these waveguide dimensions using a finite-difference eigenmode solver (Ansys Lumerical MODE [2]). The simulated electric field intensity is plotted in Fig. 1.

For the TE mode at $\lambda = 1550$ nm, the simulated effective index is $n_{\mathrm{eff}} = 2.4446$, the group index is $n_{\mathrm{g}} = 4.1967$, and the TE polarization fraction is 0.98, meaning that it is almost entirely TE polarized. The simulated loss is 0.0004 dB/cm,

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This report was written in partial fulfilment of the requirements for the *UBCx: Silicon Photonics Design, Fabrication and Data Analysis* course taught by Prof. Lukas Chrostowski at the University of British Columbia (UBC). This course was administered through edX.org.

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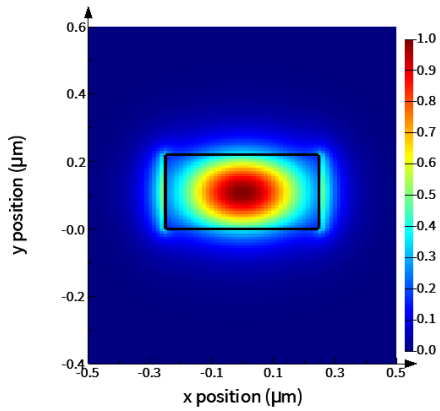

Fig. 1. The simulated electric field intensity at $\lambda = 1550$ nm. These values were simulated using a finite-difference eigenmode solver (Ansys Lumerical MODE [2]). Note that this plot is cropped to fit the waveguide. The total simulation area was 2.5 µm × 1.7 µm.

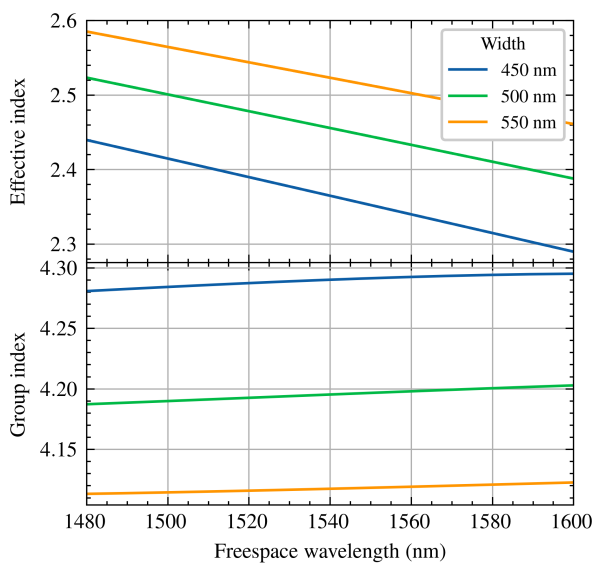

Fig. 2. The simulated propagation properties of the TE mode. These values were simulated using a finite-difference eigenmode solver (Ansys Lumerical MODE [2]). The waveguide height was set to 220 nm.

but the simulation does not include the surface roughness of the sidewalls, which is the dominant source of loss in real physical waveguides.

The simulated effective index and group index are plotted in Fig. 2 as a function of wavelength for three different waveguide widths: 450 nm, 500 nm and 550 nm. Note that the effective index rises with increasing waveguide width, suggesting that more of the energy is contained within the silicon waveguide.

I fit the simulated effective index data with the following second-order polynomial model:

$$n_{\text{eff}}(\lambda) \approx n_1 + n_2 \cdot (\lambda - 1.55) + n_3 \cdot (\lambda - 1.55)^2 \quad (1)$$

where λ is the wavelength in units of µm. This is the Taylor series expansion about $\lambda = 1.55$ µm, and is referred to as a *compact model* of the effective index. The fitted coefficient values are provided in Table II. The compact models are also plotted in Fig. 2 using dashed lines of the same colour, but they are not visible since they are in excellent agreement with the simulated values for n_{eff}.

TABLE II
COMPACT MODEL FOR THE EFFECTIVE INDEX OF THE TE MODE.

W (nm)	n_1 (1)	n_2 (µm^{-1})	n_3 (µm^{-2})
450	2.35244	-1.25065	-0.04055
500	2.44457	-1.13035	-0.04296
550	2.51302	-1.03573	-0.02595

The group index is related to the effective index of the waveguide by:

$$n_{\text{g}}(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{\text{d}n_{\text{eff}}}{\text{d}\lambda} \quad (2)$$

If we insert Eqn. 1 into Eqn. 2, we find:

$$n_{\text{g}}(\lambda) = n_1 + n_2\ (\Delta\lambda - \lambda) + n_3\ \left(\Delta\lambda^2 - 2\lambda\Delta\lambda\right) \quad (3)$$

where $\Delta\lambda = \lambda - 1.55$. At $\lambda = 1.55$ µm ($\Delta\lambda = 0$), this reduces to $n_g = n_1 - n_2 \cdot 1.55$.

From the effective index and group index, we can calculate the phase and group velocities by:

$$v_{\text{p}}(\lambda) = \frac{c}{n_{\text{eff}}(\lambda)} \quad (4)$$

$$v_{\text{g}}(\lambda) = \frac{c}{n_{\text{g}}(\lambda)} \quad (5)$$

where c is the speed of light in a vacuum. Note that silicon waveguides are dispersive, meaning that different frequencies travel at different velocities. Since the group velocity is slower than the phase velocity, the phase front appears to travel faster than the pulse envelope, as can be seen in time-domain simulations.

A. Corner Analysis

It is important to simulate the propagation properties of the waveguide to properly design the PICs. However, the dimensions of the manufactured waveguides will not exactly match the intended design. To understand what range of group index values I can expect, I conducted a *corner analysis*. This involves simulating the group index for the minimum and maximum expected waveguide width, and the minimum and maximum expected waveguide height. I simulated the four corners, which gives the worse case scenario. For the waveguides produced in this course, the 6σ waveguide widths fall between 470 nm and 510 nm, and the 6σ waveguide heights fall between 215.3 nm and 223.1 nm. Using these values in the finite-difference eigenmode solver, the group index could vary between 4.16 and 4.24.

III. PIC DESIGNS

This section discusses the design of the individual photonic integrated circuits (PICs). The layout of the entire PIC is shown in Fig. 3. The four Mach-Zehnder interferometers are on the far left (all with labels beginning in *MZI*), the ring

resonator is on the far right (*RING*), and the Bragg gratings (*BRAGG*), photonic crystals (*PHC*), Fabry–Pérot interferometer (*FP*) and de-embedding circuits (*DE*) are in the middle.

The simulated results for the PICs are plotted in Fig. 4. The transmission through the circuits was simulated using Ansys INTERCONNECT [3], along with the SiEPIC process design kit (PDK) [6]. These will be discussed in more detail in the following subsections. Note that the simulated results for the photonic crystals do not include the effect of the fiber grating because I designed these from scratch and so they were not simulated using the PDK.

A. De-Embedding Circuits

Every design presented in this paper uses fiber gratings to inject or extract light from the PIC, and strip waveguides to connect the various optical devices. To model the loss of the fiber gratings, I created two de-embedding circuits. These circuits only have two fiber gratings connected together using strip waveguides. By measuring the insertion loss of these de-embedding circuits, I can then subtract the insertion loss from the measured transmission of the other designs to partially isolate the response of the device-under-test (DUT). In microwave engineering, this is known as a *thru response calibration*. It is easy to implement, but it does not account for reflections within the PIC (e.g., reflections between the fiber grating and the DUT circuit).

I created two simple designs, as shown in Fig. 3: one with a total waveguide length of 133.9 µm (*DE1*), and another with a length of 559.3 µm (*DE2*). The second design has 22 right-angle bends, which have higher attenuation than straight waveguides. The first design only has two bends. For the fabrication process used in this paper, I expect an attenuation constant of 1.2 dB/cm for straight waveguides and 3.0 dB/cm for waveguide bends[1].

B. Imbalanced Mach-Zehdner Interferometer

The second design group has four Mach-Zehdner interferometers (MZI). These are common optical devices. Within PICs, MZIs are used to create components like switches and filters. They are constructed using simple strip waveguides and Y-branch splitters, as well as grating couplers for the input and output signals.

The transfer function of a lossless imbalanced MZI is:

$$\frac{I_\mathrm{o}}{I_\mathrm{i}} = \frac{1}{2}\left[1 + \cos\left(\beta\,\Delta\ell\right)\right] \tag{6}$$

where I_i and I_o are the input and output intensity, respectively; β is the propagation constant and $\Delta\ell$ is the length mismatch between the two branches of the MZI.

The goal of this design was to extract the group index from the MZI transmission spectra. The distance between minima in the transmission data is known as the *free spectral range* (FSR) and it is approximately equal to:

$$\mathrm{FSR} \approx \frac{\lambda^2}{\Delta\ell\, n_\mathrm{g}} \quad [\mathrm{m}] \tag{7}$$

[1] https://www.appliednt.com/nanosoi-fabrication-service/

TABLE III
SUMMARY OF MZI DESIGN VARIATIONS.

Label	Number of minima	Target FSR (nm)	$\Delta\ell$ (µm)
MZI1A/MZI1B	10	10.0	60
MZI2	20	5.0	120
MZI3	30	3.3	180

where $\Delta\ell$ is the length mismatch between the two sides of the MZI, λ is the free-space wavelength, and n_g is the group index. Therefore, if I design an imbalanced MZI and then I measure the FSR, I can extract the group index of the strip waveguide. This is the primary goal of this project.

A target of 10 minima will give me enough samples within the measurement window to see the shape of the group index. Since the measurement bandwidth is approximately 100 nm, this means that I should target an FSR of 10 nm. For a measurement centred at 1550 nm, this corresponds to a length mismatch of:

$$\Delta\ell \approx \frac{\lambda^2}{\mathrm{FSR}\cdot n_\mathrm{g}} \approx 57\ \mu\mathrm{m} \tag{8}$$

where I assumed $n_\mathrm{g} \approx 4.2$. I also created variants with 20 and 30 minima. These require length mismatches of $\Delta\ell \approx 114$ µm and $\Delta\ell \approx 172$ µm, respectively. These design variants are summarized in Table III.

These length mismatch values were validated using Ansys Lumerical INTERCONNECT [3], along with the SiEPIC E-Beam PDK [6]. The simulated transmission results through the three imbalanced MZI circuit variations are plotted in Fig. 4. For *MZ1A*, there are 10 to 11 minima, as intended.

To assess repeatability, I created two copies of the $\Delta\ell = 60$ µm design: *MZ1A* and *MZ1B*. I also created a version with $\Delta\ell = 60$ µm (*MZ2*) and a version with $\Delta\ell = 180$ µm (*MZ3*).

C. Bragg Grating

Bragg gratings are strip waveguides where the width of the waveguide is periodically modulated to create a 1-D photonic crystal. The first Bragg grating circuit (*BRAGG1* in Fig. 3) has a single Bragg grating. The width of the waveguide alternates between 550 nm and 450 nm. The length of each period is 317 nm and there are 100 periods. The stopband is centered at the Bragg wavelength:

$$\lambda_\mathrm{Bragg} = 2\, n_\mathrm{eff}\, \Lambda \tag{9}$$

where Λ is the grating period. At each discontinuity, where the waveguide changes width, there will be a reflection. At the Bragg wavelength, these reflections will combine constructively to produce a strong reflection, and thereby a stopband in the transmission response. Assuming $n_\mathrm{eff} \approx 2.44$, this should occur at approximately 1547 nm.

The simulated results for *BRAGG1* are shown in Fig. 4. A visible stopband is seen between approximately 1535 nm and 1555 nm, as intended. (For this circuit, I used the supplied process design kit to create the Bragg grating. I did not simulate or design the Bragg grating from scratch.)

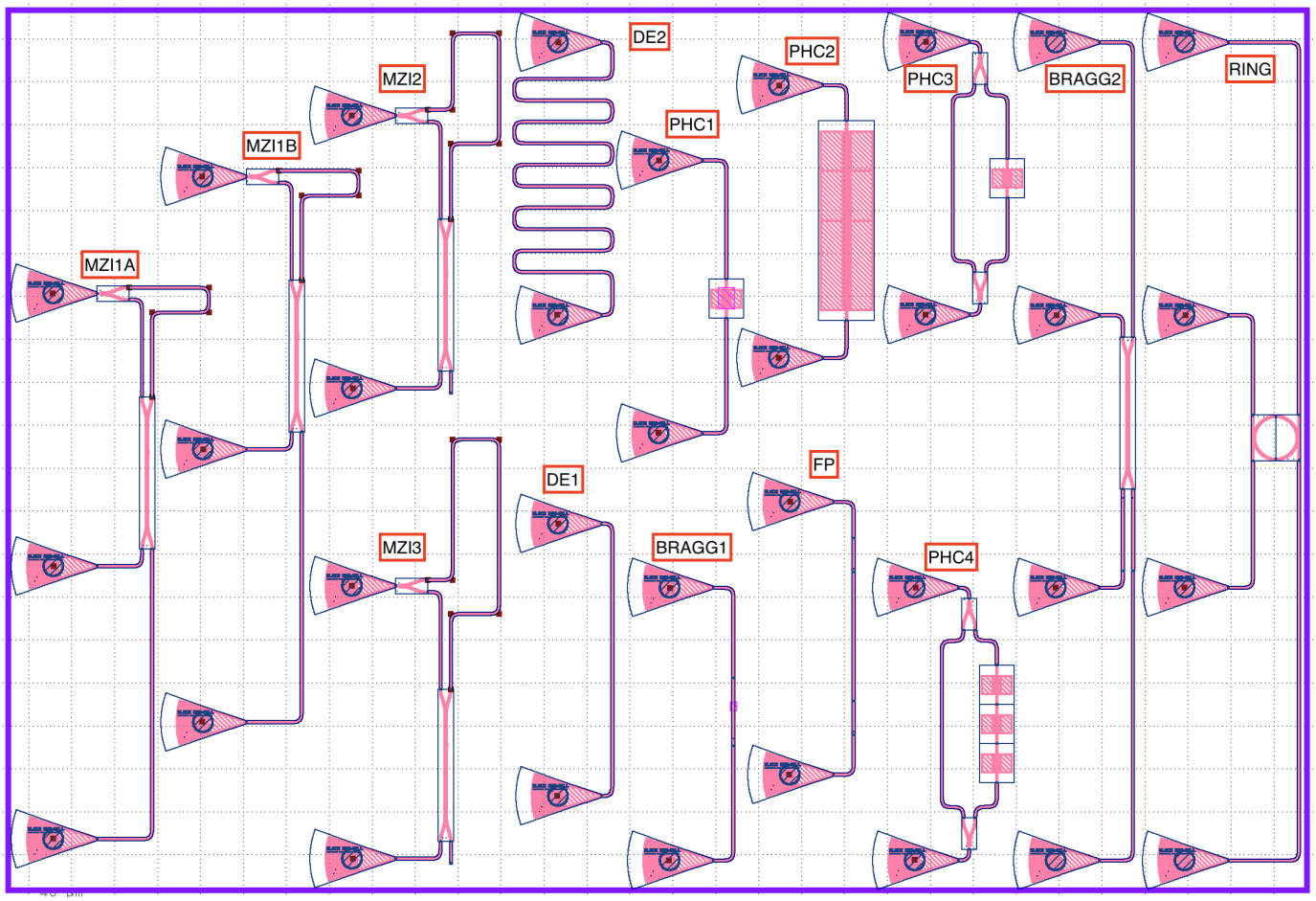

Fig. 3. Layout of my sector within the larger multi-project wafer with various PIC designs. Each student is allowed an area of 605 µm × 410 µm, which is represented by the purple box around the outside.

Since a Bragg grating is essentially a stopband filter (also known as a notch filter), I also wanted to try creating a bandpass filter using two Bragg gratings. To do this, I took a 50/50% broadband splitter, and placed a Bragg grating on each output leg of the splitter. These Bragg gratings are the same as *BRAGG1*, but they use 120 periods instead of 100. In the "stopband" of the Bragg gratings, incident waves should reflect off the Bragg gratings, recombine through the splitter, and then appear at the output port (the top fiber grating). The simulated results for this circuit are shown in Fig. 4 in the *BRAGG2* subplot. The blue trace is the top fiber grating, showing the intended bandpass filter response.

D. Fabry-Pérot Interferometer

A Fabry-Pérot interferometer is created by two beam splitters (or any partially reflective components) separated by some distance. Light will reflect back and forth inside the cavity. There will be a peak in transmission through the cavity whenever the round-trip phase is an integer number of 2π.

For the "beam splitters", I used two Bragg gratings (the same Bragg gratings as *BRAGG1* except using 50 periods instead of 100). I separated them by 60 µm. Therefore, we should see a peak in transmission whenever the following condition is met:

$$2\, n_{\text{eff}}\, \ell = N\lambda \tag{10}$$

where ℓ is the distance between the two Bragg gratings and N is any integer. The free spectral range is given by:

$$\text{FSR} = \frac{\lambda^2}{2\, n_{\text{g}}\, \ell} \tag{11}$$

For $\ell = 60$ µm and assuming $n_{\text{g}} \approx 4.15$, I should expect an FSR of approximately 5 µm. The simulated results are shown in Fig. 4 with the label *FP*. The FSR is approximately 5 µm, as expected.

E. 2-D Photonic Crystals

Inside crystals, light moves in different directions at different speeds. This is described by a band diagram. We can mimick this behaviour by creating 2-D periodic structures within the PIC.

I started with an example from the Ansys website [9], which is itself based on [10]. I scaled the spacing of the holes so that the transition between the passband and stopband moved to approximately 1550 nm. This occurred when the hole spacing was set to $a = 0.415$ µm, the small holes were set to a radius

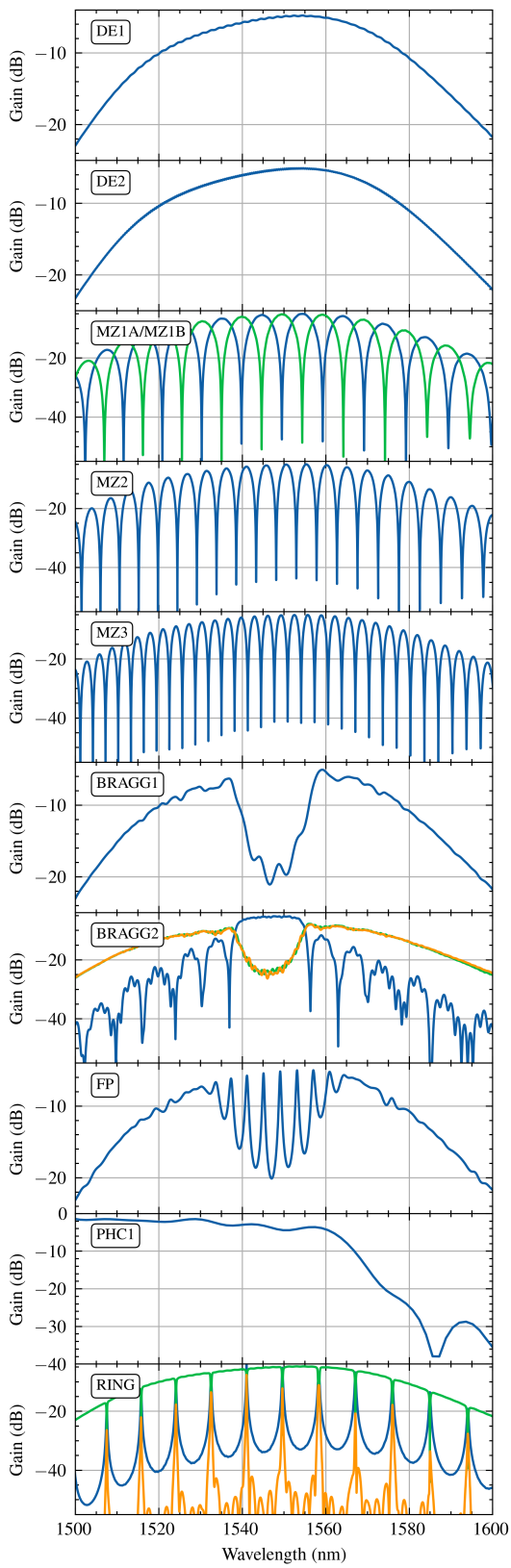

Fig. 4. Simulated transmission through the various PICs. *DE1* and *DE2* are de-embedding circuits. *MZ1A*, *MZ1B*, *MZ2* and *MZ3* are imbalanced Mach-Zehdner interferometers. *BRAGG1* and *BRAGG2* use Bragg gratings. *FP* is a Fabry–Pérot interferometer made using Bragg gratings. Finally, *RING* is a ring resonator.

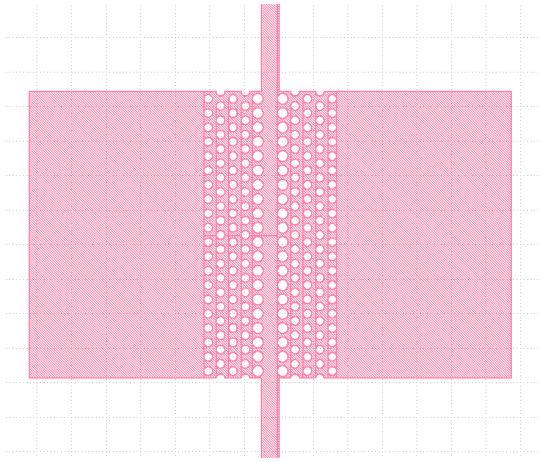

Fig. 5. Layout of the first photonic crystal (*PHC1*), as seen in KLayout. The GDS file was created using a Python script and the GDSFactory library [11].

of $r_1 = 0.125$ µm, and the large holes were set to a radius of $r_2 = 0.158$ µm.

To make the GDS file, I recreated this design using a Python script and the GDSFactory Python library [11]. Finally, I imported the GDS file into Ansys varFDTD [4] to simulate the photonic crystal and validate the design.

The layout of the photonic crystal is shown in Fig. 5 and the simulated results from Ansys varFDTD are shown in Fig. 4 with the *PHC1* label. I also made another 2-D photonic crystal, except much longer, as seen in Fig. 3 with the *PHC2* label.

F. Microring Resonator

Microring resonators are common components in PICs, used to create various components such as filters, modulators, switches and sensors. They consist of a waveguide ring weekly coupled to one or more waveguides. When the length of the waveguide ring is an integer number of wavelengths, energy can couple into the ring and then into a second waveguide. This creates sharp transmission peaks whenever the wavelength meets this condition.

In my design, the radius of the ring waveguide is $R = 10$ µm. Assuming $n_g \approx 4.2$, the free spectral range is:

$$\mathrm{FSR} \approx \frac{\lambda^2}{2\pi n_g R} \approx 9.1 \text{ nm} \tag{12}$$

The simulated results are shown in Fig. 4, and the FSR is indeed approximately 9 nm, as intended.

IV. Fabrication and Testing

The PICs presented in this paper were fabricated and tested through the openEBL[2] service offered by the Silicon Electronics-Photonics Integrated Circuits[3] (SiEPIC) program [5], [6]. For the February 2026 fabrication run, the PICs were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools, Inc.[4] in Edmonton, Alberta, Canada.

[2] https://siepic.ca/openebl/
[3] https://siepic.ca/
[4] http://www.appliednt.com/nanosoi

This process uses a state-of-the-art 100 keV electron-beam lithography system (*e-beam*). The typical loss for waveguides made through this process is 1.2 dB/cm for straight waveguides and 3.0 dB/cm for curved waveguides.

The process begins with a 200 mm SOI wafer. This wafer features a 220-nm-thick silicon layer on top of a 2-µm-thick buried oxide layer. All of this is on top of a 725-µm-thick silicon handle. The wafer was pre-diced into square substrates with dimensions of 25×25 mm, and lines were scribed into the substrate backside to facilitate easy separation into smaller chips once fabrication was complete.

After an initial wafer cleaning using piranha solution (3:1 H_2SO_4:H_2O_2) for 15 minutes and a water/IPA rinse, a negative resist (hydrogen silsesquioxane, HSQ) was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were then patterned using a Raith EBPG 5000+ e-beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected to account for proximity effects that result from the backscatter of electrons from the exposure of nearby features. The shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure, and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residue and/or defects. The chips were then mounted on a 4" handle wafer. The silicon not covered by the negative photoresist was then fully etched using an inductively coupled plasma (ICP) reactive-ion etching (RIE) process, using chlorine afterwards for qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and then the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality.

Finally, a 2.2-µm-thick oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Each designer through the openEBL program is allowed an area of 605 µm × 410 µm. This is enough room to fit approximately 5–10 simple photonic circuits, such as Mach-Zehdner or Fabry-Pérot interferometers.

A. Automated Testing

The openEBL program offers automated optical measurements in the C-band, between approximately 1500 and 1600 nm. These measurements were carried out using a custom-built automated test setup [6] with automated control software written in Python[5]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors were used as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fiber was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [8]. A 90° rotation was used to inject light into the TM grating couplers [6]. A polarization maintaining fibre array was used to couple light in and out of the chip[6]. These measurements were conducted at 25°C, with the option of additional measurements at other higher temperatures.

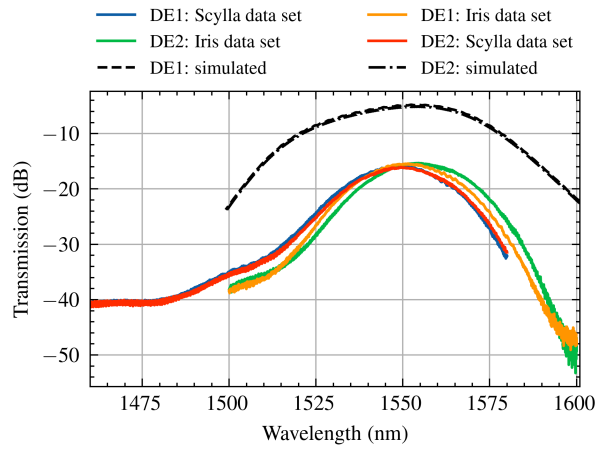

Fig. 6. Measured transmission through the two de-embedding circuits: *DE1* and *DE2*.

To work with the automated testing setup, the grating couplers (GCs) must follow a specific layout format. A maximum of four grating couplers is allowed per circuit. They must be arranged in a column with a centre-to-centre spacing of 127 µm and they must be rotated such that the waveguide exits from the right-hand side. Counting from the top GC, the second GC is the input and all of the other GCs are outputs. Note that we do not need to use all four GCs, e.g., we could choose to only use 2–4, meaning that the input is actually the top GC.

V. Measurement Results & Analysis

For each circuit, two sets of measurement results were provided by the team at SiEPIC/UBC: the first set using the measurement system described in Sec. IV-A (the "Scylla data set"), and a second set using a new measurement setup recently constructed at UBC (the "Iris data set").

A. De-Embedding Circuits

The measured transmission through the two de-embedding circuits is plotted in Fig. 6. Recall that the first de-embedding circuit has a total waveguide length of 133.9 µm and two right-angle waveguide bends, while second has a total length of 559.3 µm and 22 right-angle waveguide bends. Despite the second circuit having a significantly longer waveguide and far more waveguide bends, the results are very similar. This shows that most of the loss comes from the grating coupler.

In the following subsections, the transmission results from the first de-embedding circuit will be subtracted from the DUT to partially isolate the response of the DUT circuit.

[5] http://siepic.ubc.ca/probestation

[6] www.plcconnections.com

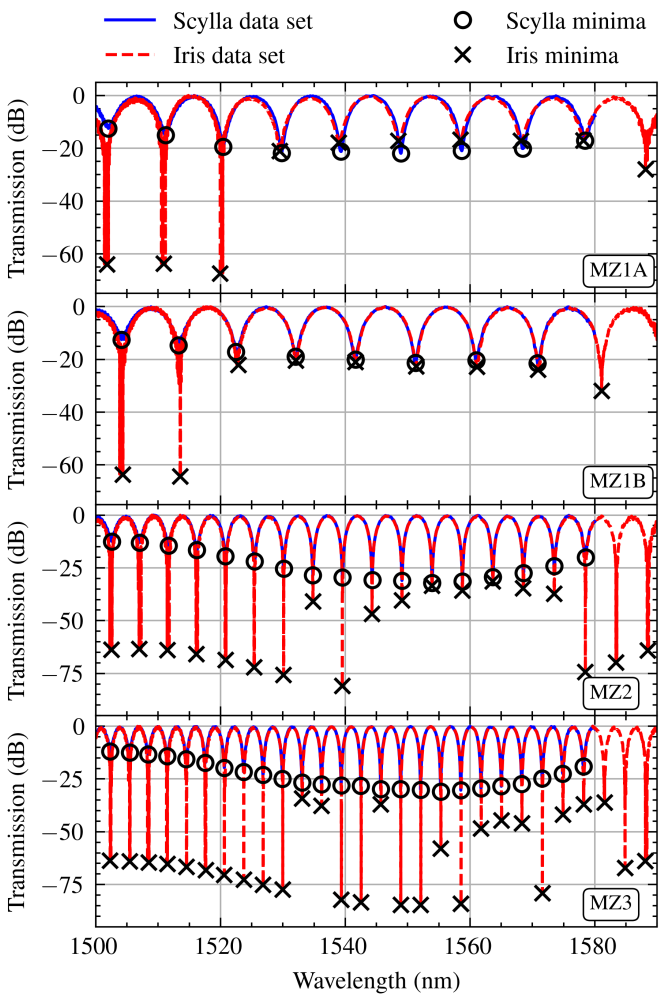

Fig. 7. Measured transmission through the four imbalanced Mach-Zehdner interferometers. MZ1A and MZI1B have a length imbalance of $\Delta\ell = 60$ µm, MZI2 has a length imbalance of $\Delta\ell = 120$ µm, and MZI3 has a length imbalance of $\Delta\ell = 180$ µm. The minima were identified using the `find_peaks` function from the Scipy library.

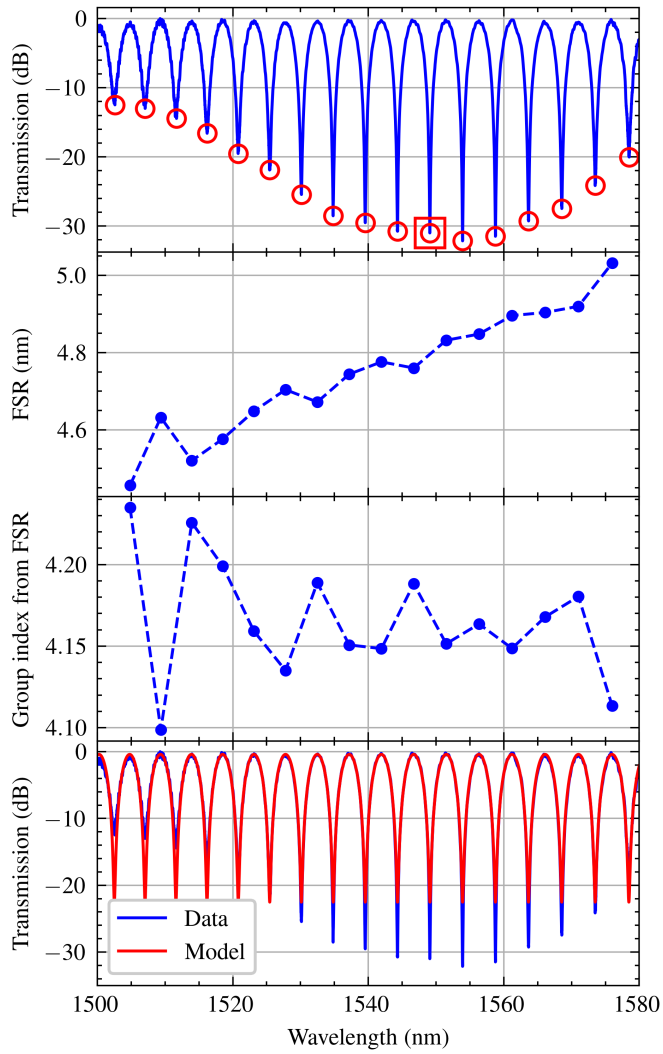

Fig. 8. Analysis of the results from the second MZI (*MZI2*).

B. Imbalanced Mach-Zehdner Interferometer

To prepare the MZI transmission data, I first truncated the data from 1500–1590 nm to remove noisy data at the highest and lowest wavelengths. I then used the `find_peaks` function from the Scipy library to find all of the peaks in transmission data. I fit a fourth-order polynomial model to the these peaks (using `polyfit` from the Numpy library), and then subtracted this polynomial from the transmission data to flatten the baseline. The results are shown in Fig. 7. For each MZI circuit, I found the minima using `find_peaks` function from the Scipy library. The minima are marked in Fig. 7.

The next steps are only shown for *MZI2* in Fig. 8, but the same process was repeated for all four circuits and with both data sets.

Using the location of the minima, I calculated the FSR, as shown in the second sub-plot in Fig. 8. I used the FSR to calculate group index by rearranging Eqn. 13:

$$n_{\mathrm{g}} \approx \frac{\lambda^2}{\Delta\ell \cdot \mathrm{FSR}} \tag{13}$$

as shown in the third sub-plot in Fig. 8. The mean group index is 4.1659, which falls within the range of acceptable values (4.16 and 4.24) that were identified through the corner analysis in Sec. II-A.

I then found the closest minima to 1550 nm, which is found at 1549.1 nm for *MZI2*. This value has a red box around it in Fig. 8. I used this value to determine an initial guess for n_1 in the compact model. First, I estimated the order of this minima using:

$$N = \mathrm{round}\left(\frac{n_1\,\Delta\ell}{\lambda} - 0.5\right) \tag{14}$$

where I assumed $n_1 = 2.44457$, which is the value that I found through finite-difference eigenmode simulations in Sec. II. I then reversed this equation to find the value of n_1 that would give me this exact order value:

$$n_1 = (2\,N + 1)\,\frac{\lambda}{2\,\Delta\ell} \tag{15}$$

This was my initial guess for n_1.

TABLE IV
SUMMARY OF GROUP INDEX RESULTS FROM ALL MZI CIRCUITS AND BOTH DATA SETS.

MZI	Scylla	Iris
MZI1A	4.1428	4.1434
MZI1B	4.1232	4.1275
MZI2	4.1599	4.1618
MZI3	4.1687	4.1701

The initial guess for n_2 was found by:

$$n_2 = \frac{-(n_\mathrm{g} - n_1)}{1550\ \mathrm{nm}} \tag{16}$$

where I assumed $n_\mathrm{g} = 4.20$.

Finally, I fit the following equation to the measured transmission data:

$$F = 10\ \log_{10}\left[\frac{1}{4}\left|1 + \exp\left(-i\frac{2\pi\, n_\mathrm{eff}}{\lambda}\Delta\ell - \frac{\alpha\Delta\ell}{2}\right)\right|^2\right] + b \tag{17}$$

where α is the waveguide loss in [Np/m], b is the excess insertion loss in [dB] and n_eff was calculated using the compact model (Eqn. 1) using the n_1 and n_2 initial guesses that I discussed above. The resulting fit is shown in the bottom sub-plot in Fig. 8. This fit was achieved with a compact model with the following terms: $n_1 = 2.4453$, $n_2 = -1.1076$ and $n_3 = 0.0264$. This corresponds to a group index of $n_\mathrm{g} = n_1 - n_2\,\lambda = 4.1621$.

This process was repeated for all of the MZI circuits, using both data sets. The results are summarized in Table IV. The extracted group indices are between 4.12 and 4.17. This is lower than the expected value of 4.20, which was found through finite-difference eigenmode simulations. About half the values fall within the range of 4.16 and 4.24, which was predicted by the corner analysis in Sec. II-A. This discrepancy could be due to the waveguide bends, or due to a slight fabrication variation. For example, for *MZI2*, if I assume that the length mismatch is 1% smaller than the intended value, the extracted group velocity changes from 4.17 to 4.20.

Overall, the results from the Scylla and Iris data sets are very similar, with a maximum discrepancy of 0.1%.

C. Bragg Grating

The results from the two Bragg grating circuits are shown in Fig. 9.

D. Fabry-Pérot Interferometer

The measured results from the Fabry-Pérot interferometer are shown in Fig. 10. I identified the peaks in the transmission spectrum using the `find_peaks` function from the Scipy library. I then used these peaks to calculate the FSR. Finally, I rewrote Eqn. 11 to calculate the group index from the FSR:

$$n_\mathrm{g} \approx \frac{\lambda^2}{2\ell \cdot \mathrm{FSR}} \tag{18}$$

where $\ell = 60$ µm is the distance between the Bragg gratings. These group index values are plotted in the bottom sub-plot in Fig. 12. The mean value is 5.0757, which is much higher

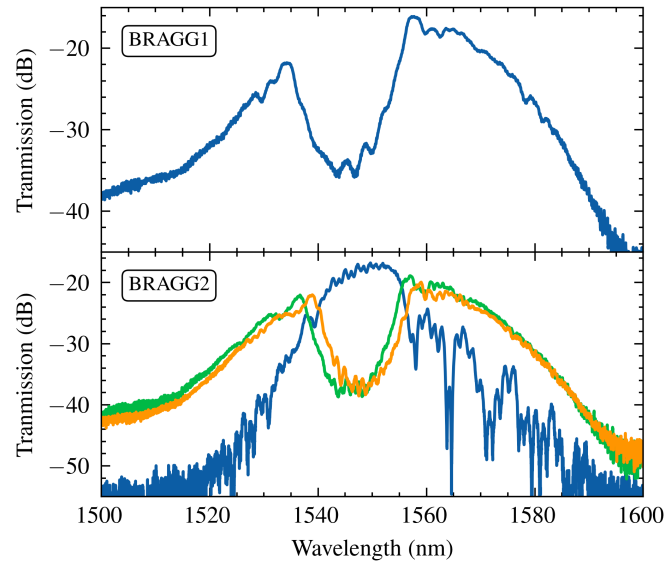

Fig. 9. Transmission results from the two Bragg grating circuits.

than the value of 4.1967 that I found using finite-difference eigenmode simulations in Sec. II. This is likely because the light penetrates some distance into the Bragg grating before reflecting. For example, if I set ℓ to 72.5 µm instead of 60 µm, the recovered group index is 4.20. Also, the group index plot has a distinct U-shape in Fig. 10. This suggests that the wave penetrates deeper into the Bragg gratings at the band edges, and penetrates less deep in the middle.

E. 2-D Photonic Crystals

A scanning electron microscope (SEM) image of the first photonic crystal is shown in Fig. 11. Unfortunately, no transmission data were collected from either of the two photonic crystals (*PHC1* or *PHC2*) in either of the two data sets. This could be due to a measurement error (e.g., misaligned optical fiber), a fabrication error (e.g., scratch through the photonic crystal), or a design error. The latter is the most likely since neither *PHC1* nor *PHC2* worked.

Using the SEM image, I measured the approximate dimensions of the photonic crystal. The hole spacing is ∼418 nm, the radius of the large holes are ∼154 nm and the radius of the small holes are ∼121 nm. These are all very close to the intended values.

F. Ring Resonator

The measured results from the ring resonator circuit are shown in Fig. 12. I identified the peaks in the transmission spectrum using the `find_peaks` function from the Scipy library. I then used these peaks to calculate the FSR. Finally, I rewrote Eqn. 12 to calculate the group index from the FSR:

$$n_\mathrm{g} \approx \frac{\lambda^2}{2\pi R \cdot \mathrm{FSR}} \tag{19}$$

These group index values are plotted in the bottom sub-plot in Fig. 12. The mean value is 4.1973, which is very close to the value of 4.1967 that I found using finite-difference eigenmode simulations in Sec. II.

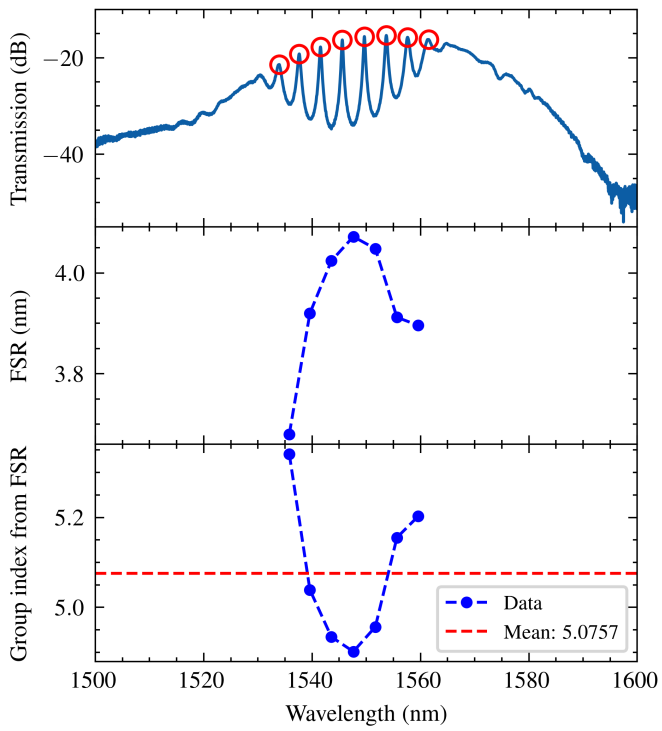

Fig. 10. Measured results from the Fabry-Pérot interferometer (FPI). The top figure shows the transmission through the FPI with the peaks labelled. The middle figure shows the free spectral range (FSR). The bottom figure shows the group index that is calculated from the FSR.

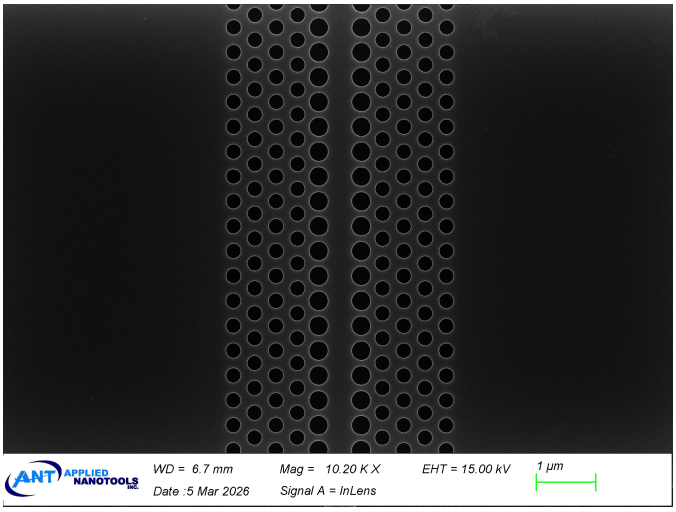

Fig. 11. Scanning electron microscope (SEM) image of the first photonic crystal (*PHC1*).

VI. Conclusion

In this paper, I presented several imbalanced Mach-Zehnder interferometers. By analyzing the transmission spectrum, I was able to recover the group index of the waveguide. My recovered group index values were between 4.12 and 4.17, which is below the value of 4.20 that I simulated using a finite-difference eigenmode solver. The discrepancy can be explained by a slight fabrication variation, or by the use of

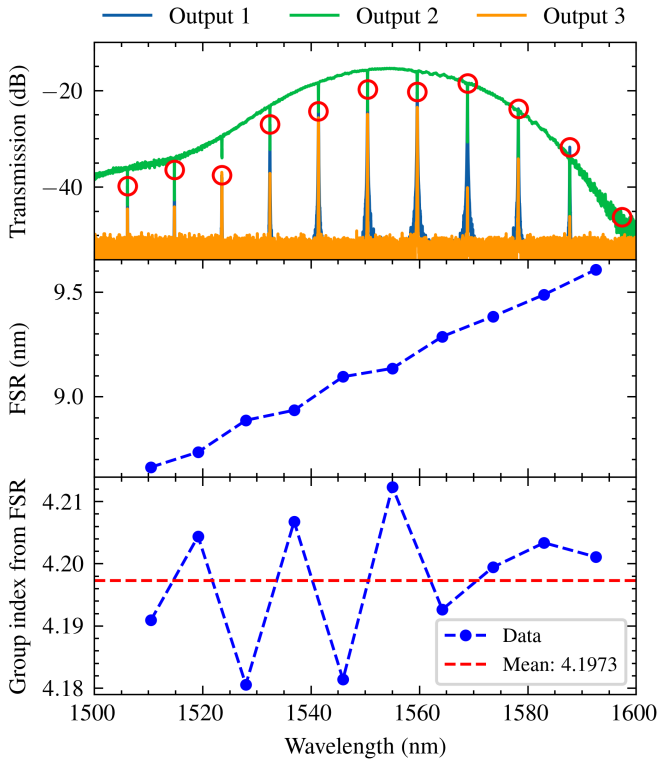

Fig. 12. Measured results from the ring resonator circuit. The top figure shows the transmission through the ring resonator with the peak labelled. The middle figure shows the free spectral range (FSR). The bottom figure shows the group index that is calculated from the FSR.

waveguide bends. I also presented several other circuits using Bragg gratings, Fabry–Pérot interferometer, ring resonators and photonic crystals.

Acknowledgment

I acknowledge the *edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis* course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. I acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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